



BITS Pilani
Pilani Campus

Machine Learning

AIML CZG565

M5 : Decision Tree Classifier

Course Faculty of M.Tech Cluster
BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Prof.Sugata, Prof.Chetana, Prof.Rajavadhana, Prof.Monali, , Prof. Srinath , Prof.Anita, Prof.Sangeetha, Prof.Swarna

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction
M2	Machine learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Module 5



- Information Theory
- Entropy Based Decision Tree Construction
- [Avoiding Overfitting
- [Minimum Description Length
- [Handling Continuous valued attributes

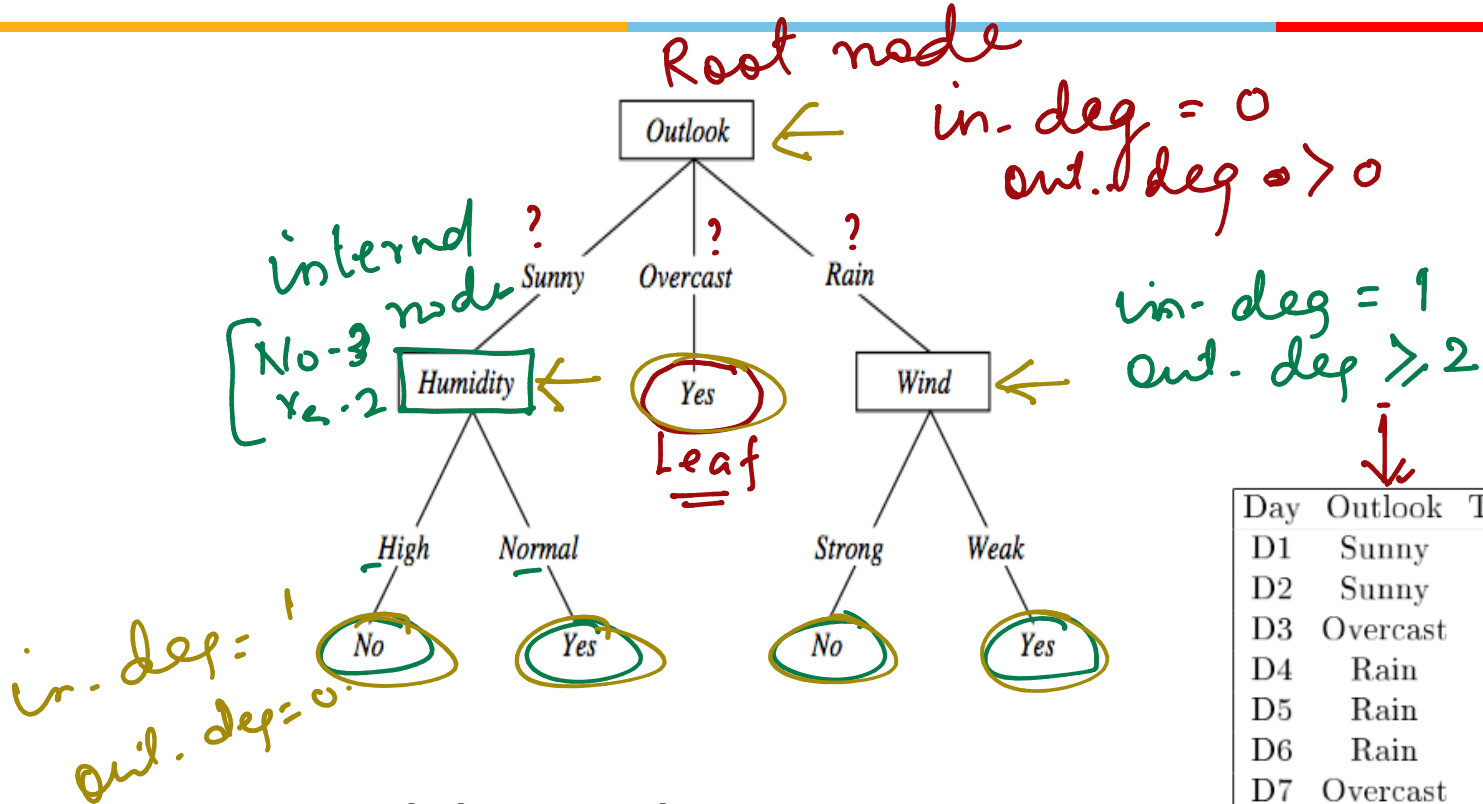
Decision Tree Classifier

Types of Classification



Decision Theory: Interpretation

Model Building



Greedy local optima.

Logistic regression, SVMs, tree based classifiers (e.g. decision tree)
Traditional neural networks, Nearest neighbor

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No ✓
D2	Sunny	Hot	High	Strong	No ✓
D3	Overcast	Hot	High	Weak	Yes —
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes —
D8	Sunny	Mild	High	Weak	No ✓
D9	Sunny	Cool	Normal	Weak	Yes ✓
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes ✓
D12	Overcast	Mild	High	Strong	Yes —
D13	Overcast	Hot	Normal	Weak	Yes —
D14	Rain	Mild	High	Strong	No

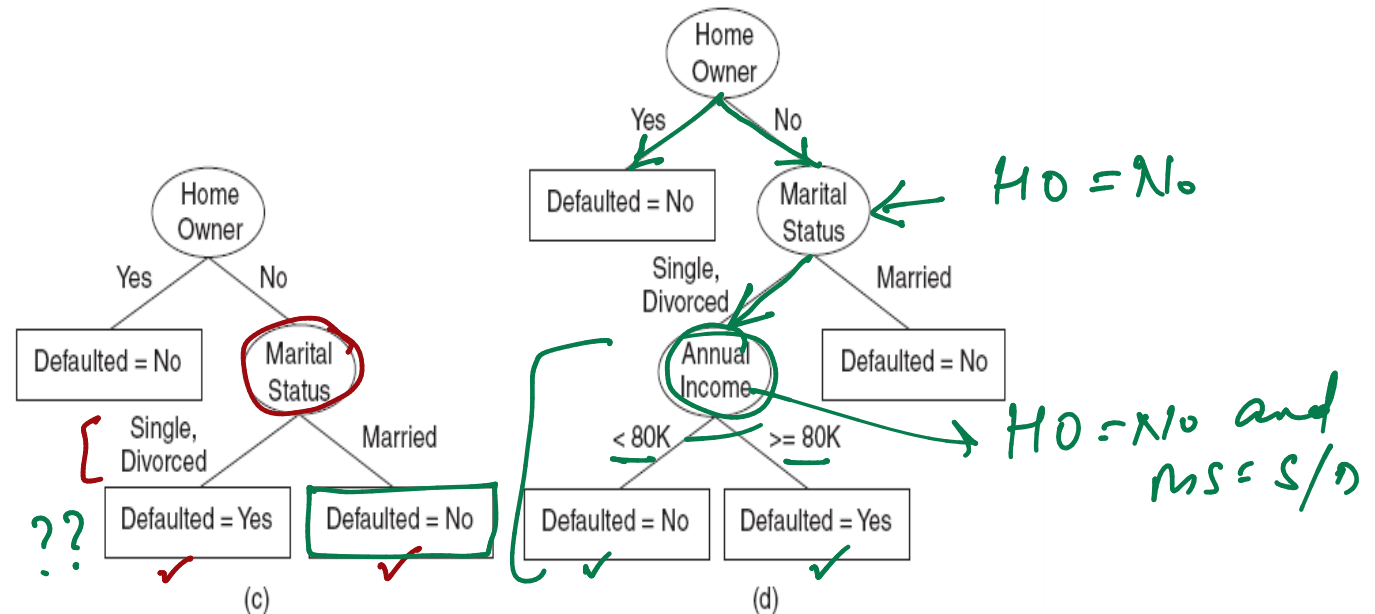
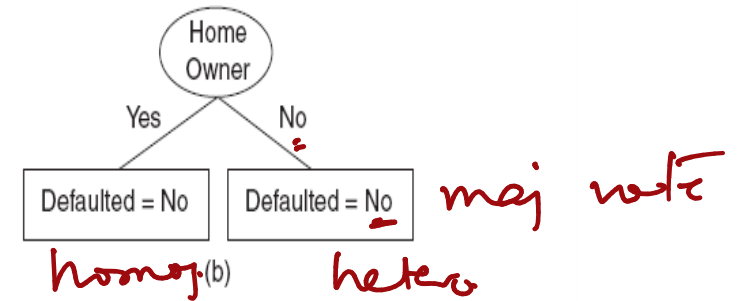
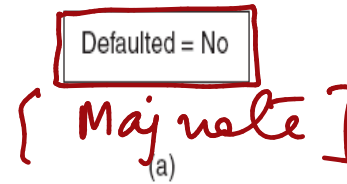
Decision Tree Construction: Hunts Algorithm



- 1) Choose a split criteria [Outlook]
- 2) If result = pure
→ leaf + stop.
else
→

Decision Tree Construction: Example

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No ✓
2	No	Married	100K	No .
3	No	Single	70K	No .
4	Yes	Married	120K	No ✓
5	No	Divorced	95K	Yes -
6	No	Married	60K	No .
7	Yes	Divorced	220K	No ✓
8	No	Single	85K	Yes .
9	No	Married	75K	No .
10	No	Single	90K	Yes -

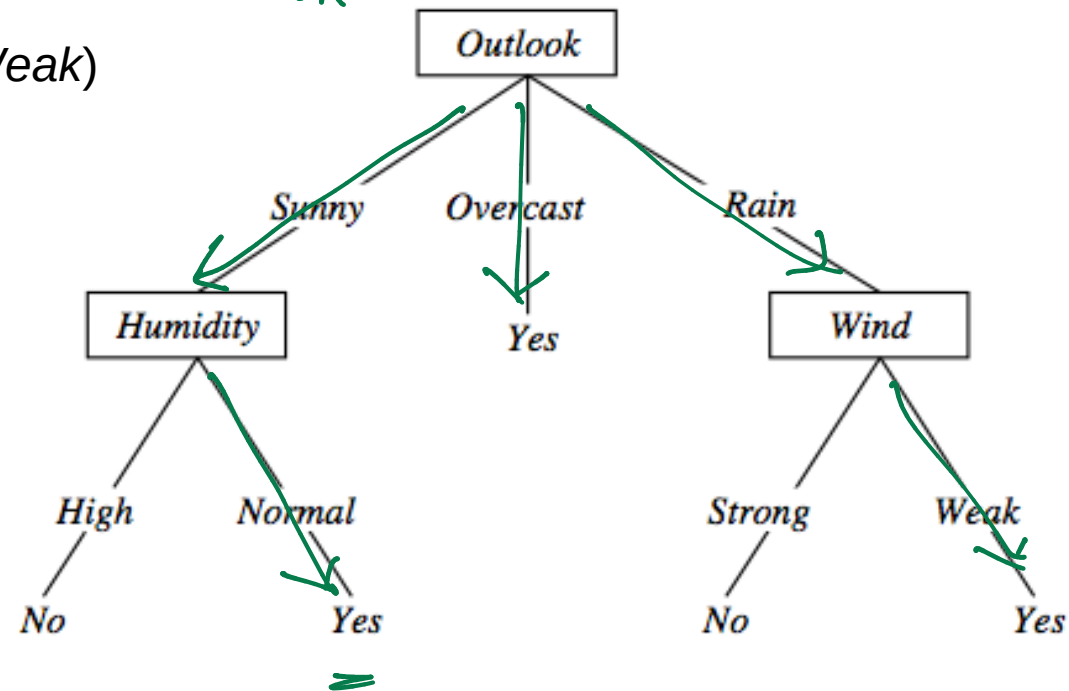


Decision Trees – Representation & Expressiveness



Decision trees represent a disjunction of conjunctions on constraints on the value of attributes: *and* *→ +ve class.*

-or
($Outlook = Sunny \wedge Humidity = Normal$) $\vee$
($Outlook = Overcast$) $\vee$
($Outlook = Rain \wedge Wind = Weak$)



=
 $\langle Outlook=Sunny, Temp=Hot, Humidity=High, Wind=Strong \rangle \rightarrow$ *Test*
No

Measure of Information

- The amount of information (surprise element) conveyed by a message is inversely proportional to its probability of occurrence. That is

$$I_k \propto \frac{1}{p_k}$$

$$I_k \propto \frac{1}{p_k}$$

- The mathematical operator satisfies above properties is the logarithmic operator. $I_k = \log_r \frac{1}{p_k} \text{ units}$

$$I_k = \log \frac{1}{p_k} = -\log p_k$$

pure ← low ← Entropy (impurity).
 impure ← high ←

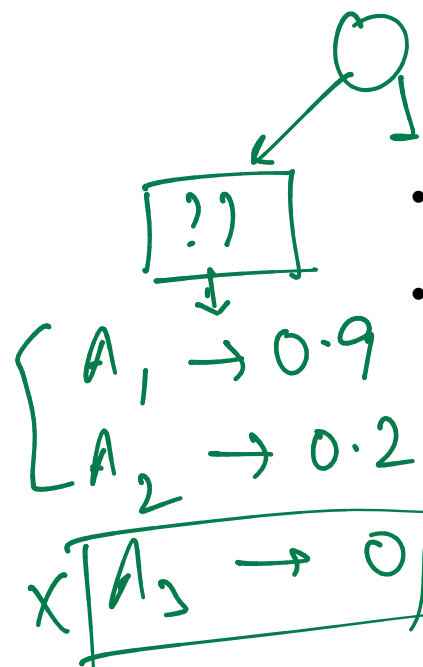
Entropy

- Entropy of discrete random variable $X=\{x_1, x_2 \dots x_n\}$

$$H(X) = E[I(X)] = E[-\log(P(X))].$$

since: $\log_2(1/P(\text{event})) = -\log_2 P(\text{event})$

- As uncertainty increases, entropy increases
- Entropy across all values



$$H(X) = - \sum_{i=1}^n P(x_i) \log_b P(x_i)$$

high - mixed
low - one class dominates
0 → pure split?

$$[X] \rightarrow x_1, x_2, x_3, \dots, x_n$$
$$P(x_1), P(x_2), \dots, P(x_n)$$

$$H(X) = E(I(X))$$
$$= E[-\log P(X)]$$

Entropy in general

- Entropy measures the amount of information in a random variable

$$H(X) = -p_+ \log_2 p_+ - p_- \log_2 p_- \quad X = \{+, -\}$$

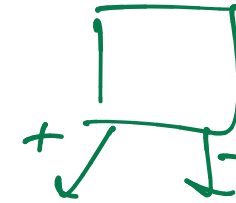
for binary classification [two-valued random variable]

$$H(X) = -\sum_{i=1}^c p_i \log_2 p_i = \sum_{i=1}^c p_i \log_2 1/p_i \quad X = \{i, \dots, c\}$$

for classification in c classes

low entropy

$$p_+ \\ p_- = 1 - p_+$$



$$[-p_+ \log p_+ - p_- \log p_-]$$

$$p_1 \quad p_2 \quad p_3 \quad \dots \quad p_c$$

$$\left[-\sum_{i=1}^c p_i \log p_i \right]$$

Entropy in binary Classification

$\log_2$

$\frac{d_+}{d_+ + d_-}$

- Entropy measures the *impurity* of a collection of examples. It depends from the distribution of the random variable p .
 - S is a collection of training examples
 - p_+ the proportion of positive examples in S
 - p_- the proportion of negative examples in S

$p_+ + p_-$

Entropy (S) $\equiv -p_+ \log_2 p_+ - p_- \log_2 p_-$ $[0 \log_2 0 = 0]$

Entropy ($[14+, 0-]$) $= -14/14 \log_2 (14/14) - 0 \log_2 (0) = 0$

Entropy ($[9+, 5-]$) $= -9/14 \log_2 (9/14) - 5/14 \log_2 (5/14) \approx 0.94$

Entropy ($[7+, 7-]$) $= -7/14 \log_2 (7/14) - 7/14 \log_2 (7/14) =$

$= 1/2 + 1/2 = 1$ $[\log_2 1/2 = -1]$

$14+$
 $p_+ = \frac{14}{14}$

$0-$
 $p_- = \frac{0}{14}$

$E(14, 0)$ $= -\frac{14}{14} \log_2 \frac{14}{14} - \frac{0}{14} \log_2 \frac{0}{14}$
 $= 0$

Note: the log of a number < 1 is negative, $0 \leq p \leq 1$, $0 \leq \text{entropy} \leq 1$

<https://www.easycalculation.com/log-base2-calculator.php>

$\log 1 = 0$
 $0 \log 0 = 0$

$\frac{9}{9+5}$

$E(9, 5)$
 $p_+ = \frac{9}{14}$

$p_- = \frac{5}{14}$

$E \rightarrow 0$ to 1
least best

Decision Trees – Problem Type 1

Entropy of the Decision

Feature EnjoySport:

Target

+ 3 - 1

[dataset]

= Entropy_{DecisionClass}(S) = $\sum p(c) \log_2 p(c)$
= $E(3,1) = - (3/4) \log_2 (3/4) - (1/4) \log_2 (1/4) = 0.4421 + 0.5 = 0.94$

E(3,1)

target
↓

Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes +
Sunny	Warm	High	Strong	Same	Yes +
Rainy	Cold	High	Strong	Change	No -
Sunny	Warm	High	Strong	Change	Yes +
Rainy	Warm	Normal	Breeze	Same	Yes

Tr.

Test

Note:

- Class balancing is recommended before building
- Value = 0 to log base 2 k, where k is the number of classes you have.

Decision Trees – Problem Type 1

$$E(0,1) = -\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1}$$

Entropy of the Decision

Feature EnjoySport:

$$E(3,0) = -\frac{3}{3} \log_2 \frac{3}{3} - \frac{0}{3} \log_2 \frac{0}{3} = 0$$

= **Entropy** _{DecisionClass} (**S**) = $\sum p(c) \log_2 p(c)$

$$= E(3,1) = - (3/4) \log_2 (3/4) - (1/4) \log_2 (1/4) = 0.4421 + 0.5 = 0.8113$$

Entropy of the attribute Sky:

$$= p(\text{Sunny}) \cdot E(3,0) + p(\text{Rainy}) \cdot E(0,1)$$

$$= (3/4) [- (3/3) \log_2 (3/3) - (0/3) \log_2 (0/3)] + (1/4) [- (0/1) \log_2 (0/1) - (1/1) \log_2 (1/1)]$$

$$= 0$$

Entropy of the attribute AirTemp:

$$= p(\text{Warm}) \cdot E(3,0) + p(\text{Cold}) \cdot E(0,1)$$

$$= (3/4) [- (3/3) \log_2 (3/3) - (0/3) \log_2 (0/3)] + (1/4) [- (0/1) \log_2 (0/1) - (1/1) \log_2 (1/1)]$$

$$= 0$$

Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	Yes
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes

Sky	Enjoy Y	Not Enjoy N
Sunny	3	0
Rainy	0	1

AirTemp	Enjoy Y	Not Enjoy N	Total
Warm	3	0	3
Cold	0	1	1

$$E(AT) = p(W) \cdot E(W) + p(C) \cdot E(C)$$

$$\rightarrow = \frac{3}{4} \cdot E(3,0) + \frac{1}{4} E(0,1)$$

(A1)

class Y/N/M

atl - a/b/c/d

	Y	N	M	
a	⊕	⊕	⊕	$P(a) \leftarrow \frac{d(a)}{d(\text{total})}$
b				$P(b)$
c				$P(c)$
d				$P(d)$

$$\rightarrow \left[E(A1) = P(a) \cdot E(a) + P(b) \cdot E(b) + \dots \right]$$
$$E(a) = -P_Y \log P_Y - P_N \log P_N - P_M \log P_M$$

calculator

Decision Trees – Problem Type 1

Entropy of the Decision
Feature EnjoySport:

Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	Yes
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes

$$= \text{Entropy}_{\text{DecisionClass}}(\mathbf{S}) = \sum -p(c) \log_2 p(c)$$

$$= E(3,1) = - (3/4) \log_2 (3/4) - (1/4) \log_2 (1/4) = 0.94$$

Entropy of the attribute Humidity : ✓

$$= p(\text{Normal}) \cdot E(1,0) + p(\text{High}) \cdot E(2,1)$$

$$= (1/4) [- (1/1) \log_2 (1/1) - (0/1) \log_2 (0/1)] + (3/4) [- (2/3) \log_2 (2/3) - (1/3) \log_2 (1/3)]$$

$$= 0 + \frac{3}{4} [0.39 + 0.52] = 0.68$$

Humidity	Enjoy	Not Enjoy
Normal	1	0
High	2	1

Entropy of the attribute Wind :

$$= p(\text{Strong}) \cdot E(3,1)$$

$$= (4/4) [- (3/4) \log_2 (3/4) - (1/4) \log_2 (1/4)]$$

$$= 1(0.94) = 0.94$$

Wind	Enjoy	Not Enjoy
Strong	3	1

Decision Trees – Problem Type 1

Entropy of the Decision
Feature EnjoySport:

Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	Yes
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes

$$= \text{Entropy}_{\text{DecisionClass}}(\mathbf{S}) = \sum -p(c) \log_2 p(c)$$

$$= E(3,1) = - (3/4) \log_2 (3/4) - (1/4) \log_2 (1/4) = 0.94$$

Entropy of the attribute Forecast :

$$= p(\text{Same}) \cdot E(2,0) + p(\text{Change}) \cdot E(1,1)$$

$$= (2/4) [- (2/2) \log_2 (2/2) - (0/2) \log_2 (0/2)] + (2/4) [- (1/2) \log_2 (1/2) - (1/2) \log_2 (1/2)]$$

$$= 0 + 0.5 = 0.5$$

Forecast	Enjoy	Not Enjoy
Same	2	0
Change	1	1

$P(s)$

$P(c)$

$$E(F) = P(s) \cdot E(2,0) + P(c) \cdot E(1,1)$$

Decision Trees – Problem Type 1

Entropy of the Decision Feature EnjoySport: 0.94

Entropy of the attribute Sky: 0

Entropy of the attribute AirTemp: 0

Entropy of the attribute Humidity: 0.68

Entropy of the attribute Wind: 0.94

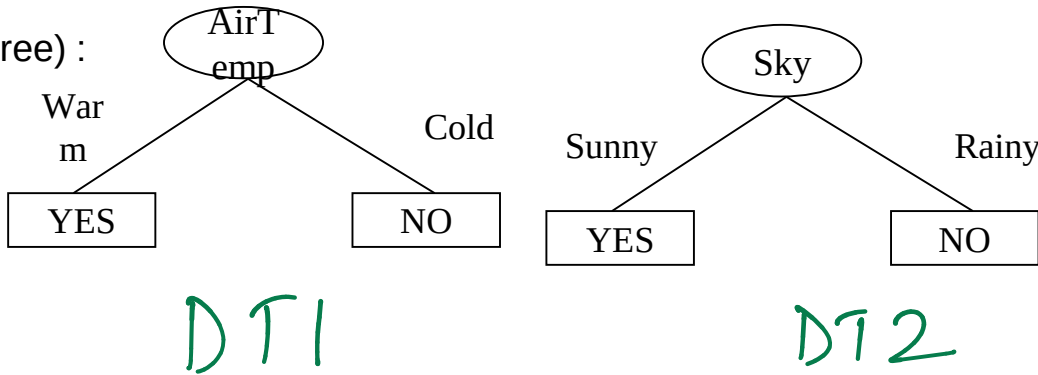
Entropy of the attribute Forecast: 0.5

Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	Yes
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes

Attribute that generalizes the model faster = That which reduces the randomness if used to decide =
That which have the highest info gain = SKY & AIRTEMP

Decision Tree constructed

(both are equally correct decision tree) :



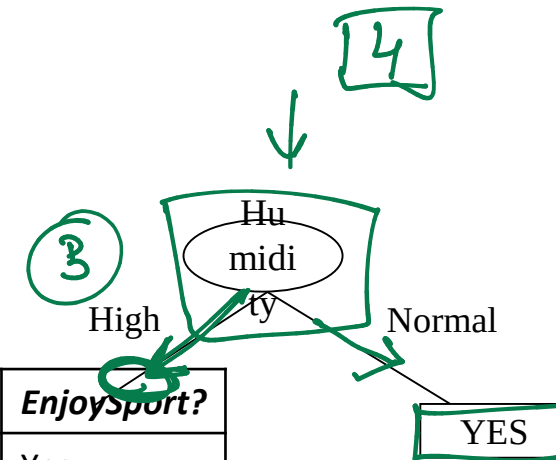
Decision Trees – Problem Type 2



(If there are heterogeneous nodes, continue the decision tree construction)

Sub Problem which has more scope to Model: None

Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes ✓
Sunny	Warm	High	Strong	Same	Yes
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes



Assumed sample data in the node

Sky	AirTemp	Wind	Forecast	EnjoySport?
Sunny	Warm	Strong	Same	Yes
Rainy	Cold	Strong	Change	No
Sunny	Warm	Strong	Change	Yes

Convergence Criteria :

Repeat the step again for the remaining features till:

- All training data are learnt
- No more learning is possible
- Constraints posed by the hyperparameter satisfied (More on this will be discussed in the next session under handling overfitting)

Information gain as entropy reduction

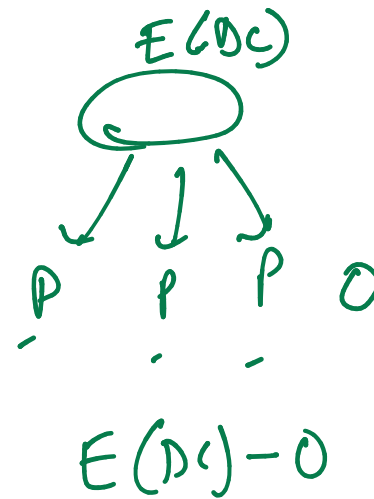
- *Information gain* is the expected reduction in entropy caused by partitioning the examples on an attribute.
- The higher the information gain the more effective the attribute in classifying training data.
- Expected reduction in entropy knowing A

$$Gain(S, A) = Entropy(S) - \left[\sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) \right]$$

$Values(A)$ possible values for A

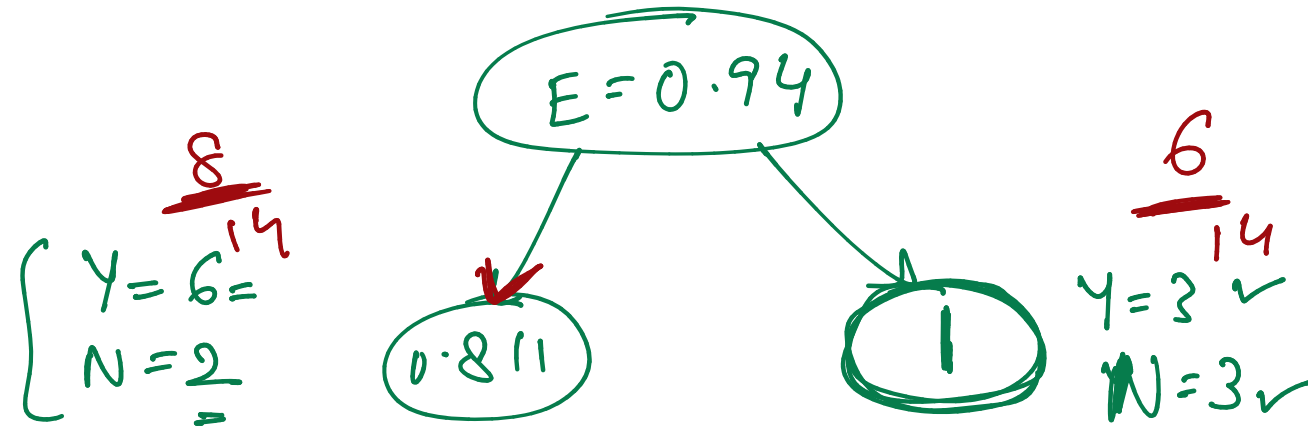
S_v subset of S for which A has value v

Maximizing the gain = minimizing the weighted average impurity measure of children nodes



$$(8, 0) = 0$$

Entropy $\rightarrow$ lowest
Gain $\rightarrow$ highest



$\rightarrow$ total = 14

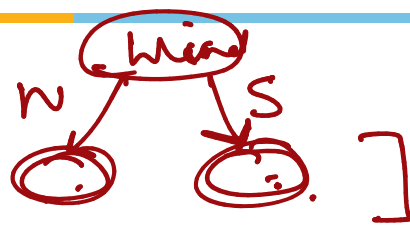
$E(6,2) = 0.8113$
 $-\frac{3}{6} \log \frac{3}{6} - \frac{2}{6} \log \frac{2}{6} = 1$
 $E(attr)$
 $Gain = \frac{0.94}{E(p)} - \left[\frac{8}{14} (0.8113) + \frac{6}{14} (1.00) \right]$

Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$\begin{aligned}
 E(\text{dataset}) &= E(9,5) \\
 &= 0.94
 \end{aligned}$$

Example: Information gain

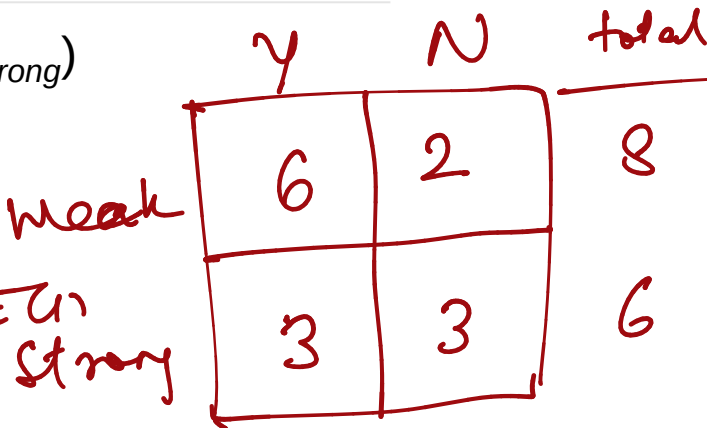
- Let 
 - $Values(Wind) = \{Weak, Strong\}$
 - $S = [9+, 5-]$
 - $S_{Weak} = [6+, 2-]$
 - $S_{Strong} = [3+, 3-]$
- Information gain due to knowing *Wind*:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$\begin{aligned}
 \text{Gain}(S, \text{Wind}) &= \text{Entropy}(S) - \frac{8}{14} \text{Entropy}(S_{Weak}) - \frac{6}{14} \text{Entropy}(S_{Strong}) \\
 &= 0.94 - \frac{8}{14} \times 0.811 - \frac{6}{14} \times 1.00 \\
 &= 0.048
 \end{aligned}$$

$$\begin{aligned}
 E(Weak) &= E(6, 2) \checkmark \\
 E(S) &= E(3, 3) \checkmark
 \end{aligned}$$

$$\begin{aligned}
 E(Wind) &= P(W) E(W) + P(S) E(S) \\
 E(P) &= E(Child)
 \end{aligned}$$



	Y	N	total
Weak	6	2	8
Strong	3	3	6

$$E(\text{Parent})$$

$$E(\text{attr})$$

$$\text{Gain} = \underline{E(P)} - \underset{\text{Wind}}{E(\underline{\text{attr}})}$$

$$\text{Inf. Gain} = E(S) - \left[\frac{8}{15} (0.871) + \frac{6}{15} (1) \right] =$$

Humidity

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



$$E(P) = E(9, 5)$$

$$E(\text{Humidity}) =$$

$$= \left[P(H) \cdot E(H) + P(N) \cdot E(N) \right]$$

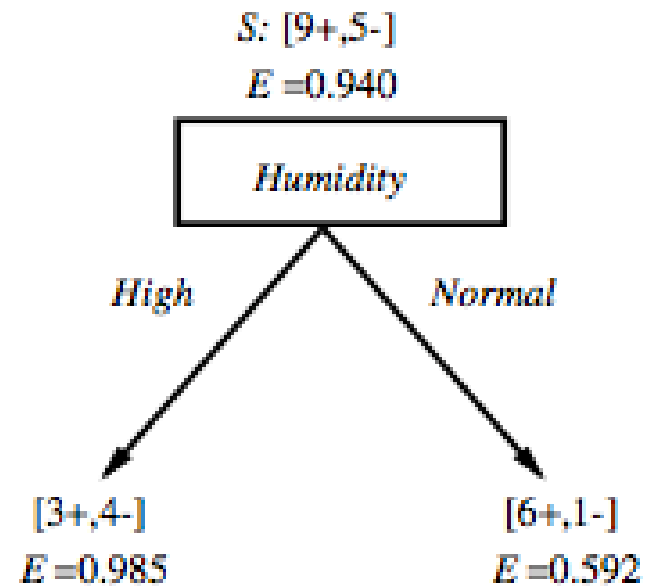
$$= \left[\frac{7}{14} \left(\underline{E(3, 4)} \right) + \frac{7}{14} \left(\underline{E(6, 1)} \right) \right]$$

	Y	N	total
H	3	4	7
N	6	1	7

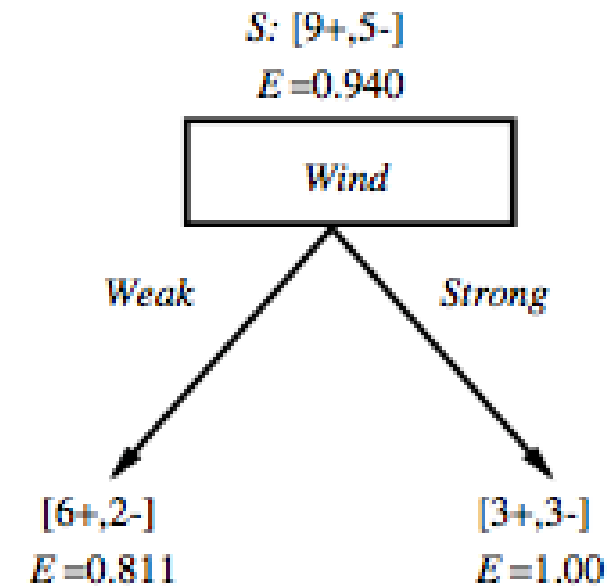
$$IG = E(9, 5) - E(\text{Humidity})$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
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D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Which attribute is the best classifier?



$$\begin{aligned} \text{Gain (S, Humidity)} &= .940 - (7/14).985 - (7/14).592 \\ &= .151 \end{aligned}$$

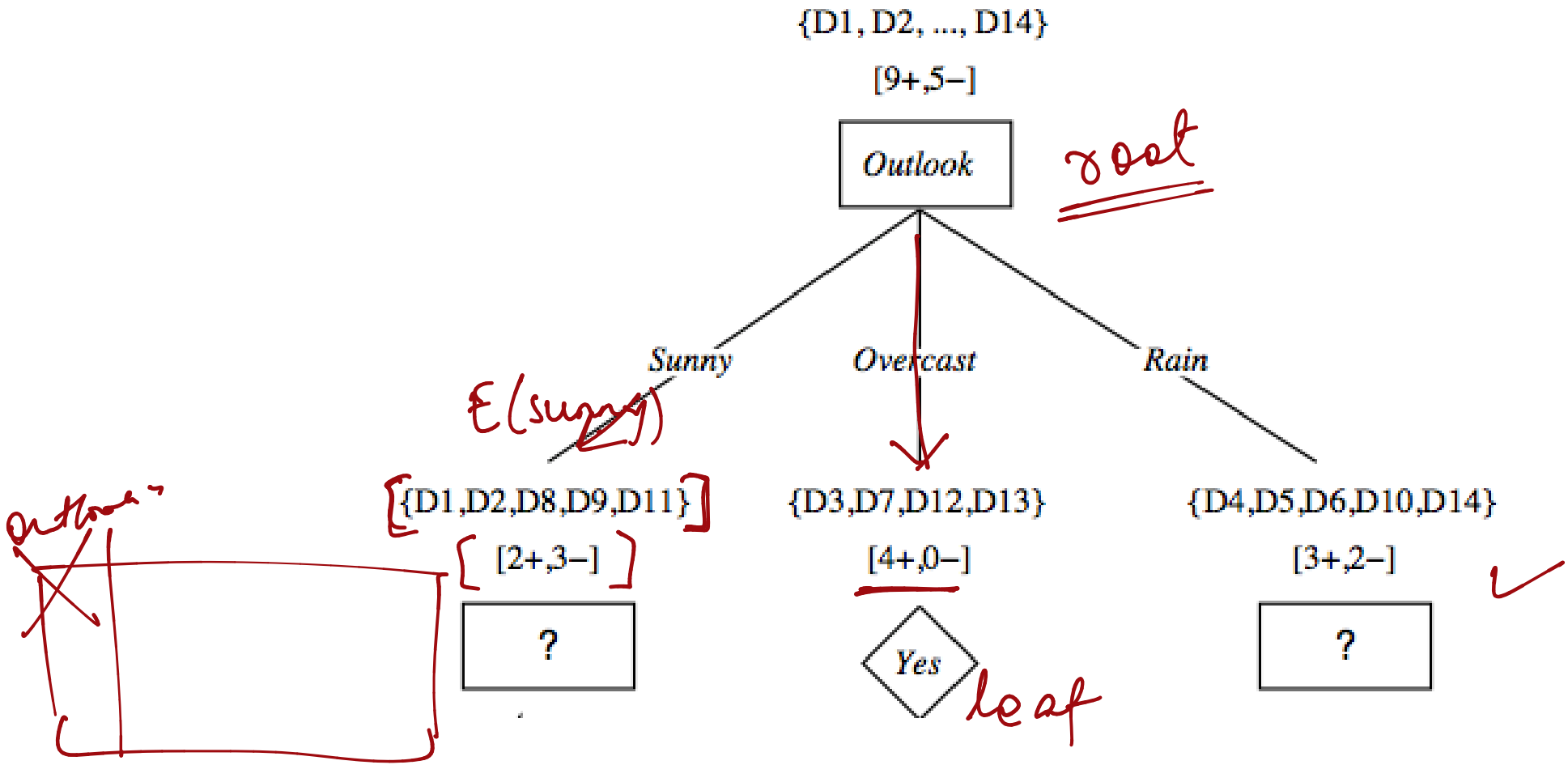


$$\begin{aligned} \text{Gain}(S, \text{Wind}) \\ &= .940 - (8/14).811 - (6/14)1.0 \\ &= .048 \end{aligned}$$

First step: which attribute to test at the root?

- Which attribute should be tested at the root?
 - $Gain(S, Outlook) = 0.246$
 - $Gain(S, Humidity) = 0.151$
 - $Gain(S, Wind) = 0.084$
 - $Gain(S, Temperature) = 0.029$
- **Outlook** provides the best prediction for the target
- Lets grow the tree:
 - add to the tree a successor for each possible value of *Outlook*
 - partition the training samples according to the value of *Outlook*

After first step



Second step



- Working on *Outlook=Sunny* node:

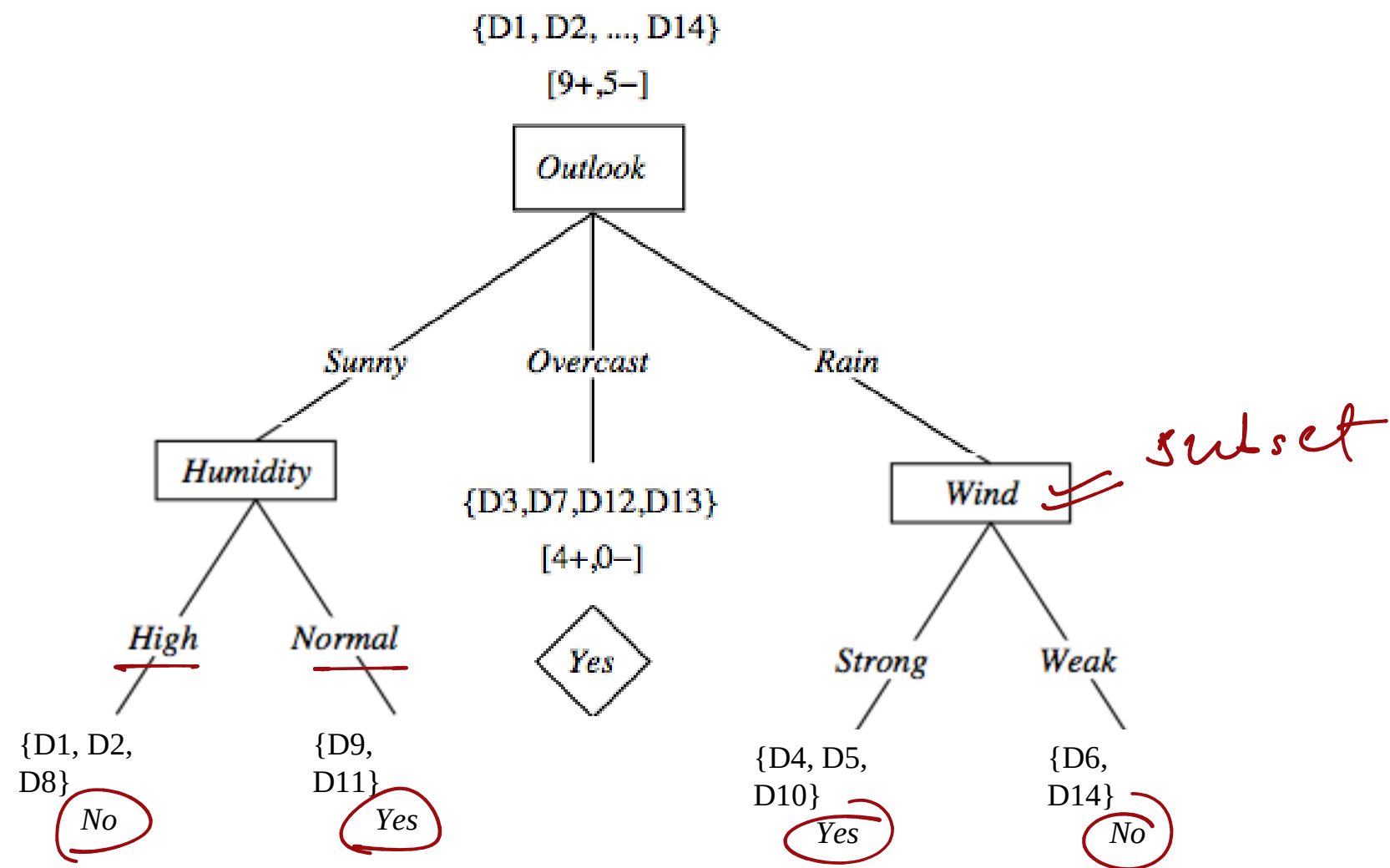
$$Gain(S_{Sunny}, \text{Humidity}) = 0.970 - [3/5 \times 0.0 - 2/5 \times 0.0] = \mathbf{0.970}$$

$$Gain(S_{Sunny}, \text{Wind}) = 0.970 - 2/5 \times 1.0 - 3.5 \times 0.918 = 0.019$$

$$Gain(S_{Sunny}, \text{Temp.}) = 0.970 - 2/5 \times 0.0 - 2/5 \times 1.0 - 1/5 \times 0.0 = 0.570$$

- Humidity** provides the best prediction for the target
- Lets grow the tree:
 - add to the tree a successor for each possible value of *Humidity*
 - partition the training samples according to the value of *Humidity*

Second and third steps



Convergence of Algorithm

- Conditions for stopping partitioning

- ✓ All samples for a given node belong to the same class
- ✓ There are no remaining attributes for further partitioning – majority voting is employed for classifying the leaf
- ✓ There are no samples left

$$\frac{GD}{\Delta \theta} \leq \epsilon$$

ID3: algorithm -Summary



ID3(X , T , $Attrs$) X : training examples:
 T : target attribute (e.g. *PlayTennis*),
 $Attrs$: other attributes, initially all attributes

Create *Root* node

If all X 's are +, *return* *Root* with class +

If all X 's are –, *return* *Root* with class –

If $Attrs$ is empty *return* *Root* with class most common value of T in X

else

$A \leftarrow$ best attribute; decision attribute for *Root* $\leftarrow A$

For each possible value v_i of A :

- add a new branch below *Root*, for test $A = v_i$

- $X_i \leftarrow$ subset of X with $A = v_i$

- *If* X_i is empty *then* add a new leaf with class the most common value of T in X

 - else* add the subtree generated by ID3(X_i , T , $Attrs - \{A\}$)

return *Root*

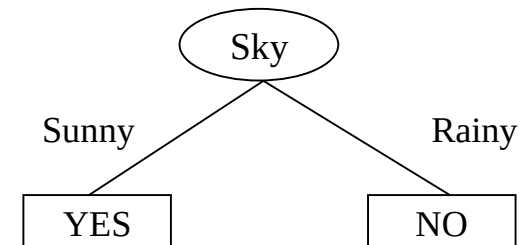
Decision Trees – Problem Type 3



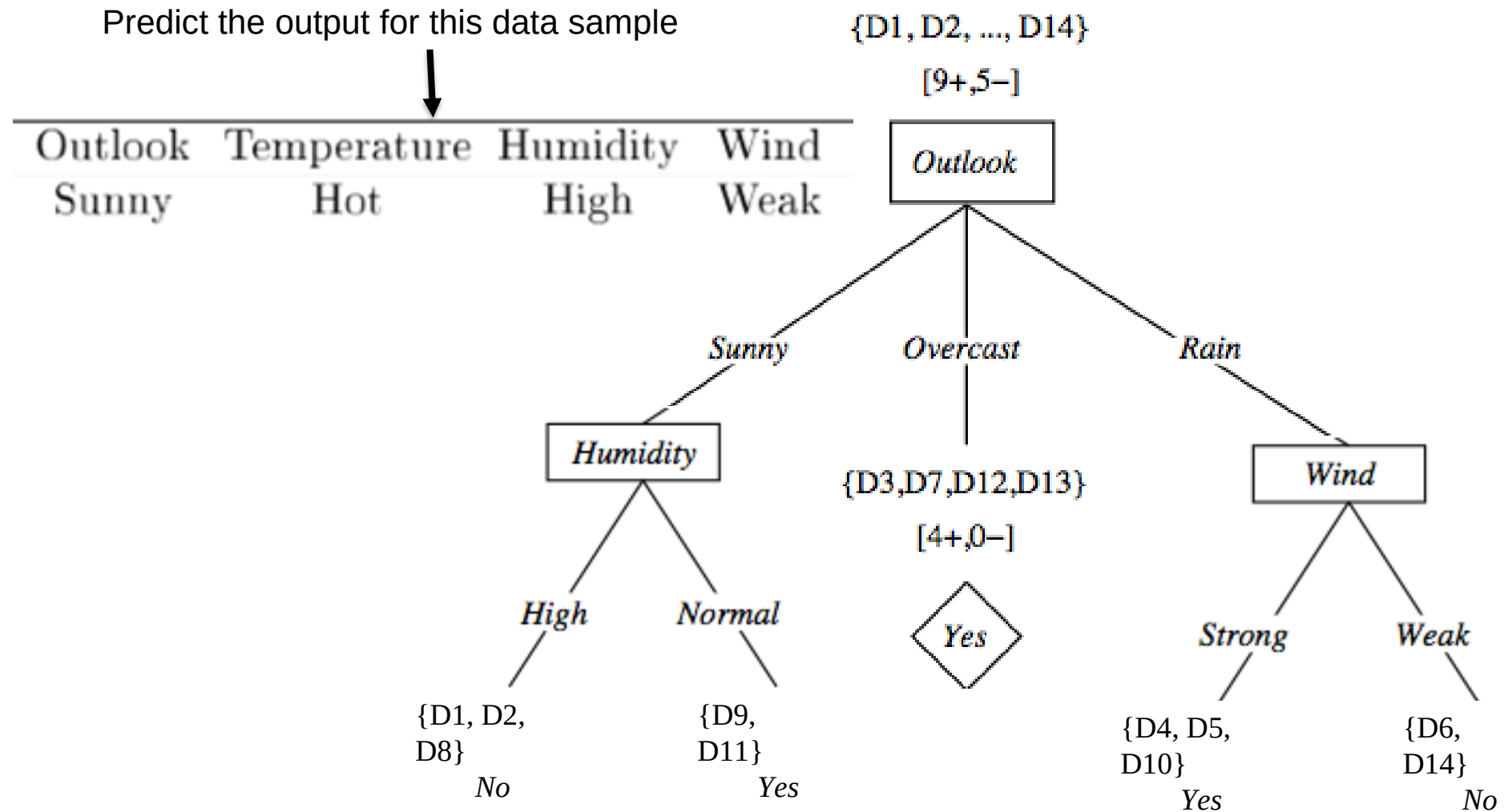
Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	Yes
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	????

Predict the class for new input instance ☐

No.



Decision Trees – Problem Type 3



Problem Type 4 – Evaluate the model

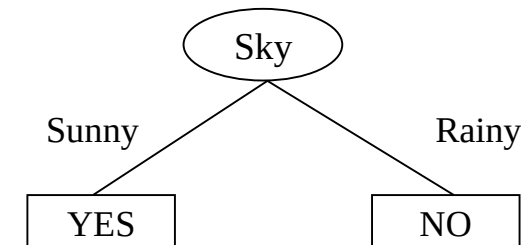
Confusion Matrix

- The evaluation discussed under module 4 is applicable here
- By default , model's predictive quality is evaluated **only by the unseen validation/test instances.**
- When training performance (eg., accuracy) is required for comparison, the training instances are used for assessment

Actual class\Predicted class	C_1	$\neg C_1$
	True Positives (TP)	False Negatives (FN)
$\neg C_1$	False Positives (FP)	True Negatives (TN)

- $CM_{i,j}$ indicates the number of tuples of class i that were labelled by the classifier as class j .

Sky	AirTemp	Humidity	Wind	Forecast	EnjoySport ?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	Yes
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	Yes



Decision trees

- Advantages:
 - Inexpensive to construct ✓
 - Extremely fast at classifying unknown records ✓ →
 - ✓ – Easy to interpret for small-sized trees
 - ✓ – Can easily handle redundant or irrelevant attributes (unless the attributes are interacting)
- Disadvantages:
 - Space of possible decision trees is exponentially large.
 - Greedy approaches are often unable to find the best tree.
 - Does not take into account interactions between attributes
 - Each decision boundary involves only a single attribute] x

Applications



Suited for following classification problems:

- Applications whose Instances are represented by attribute-value pairs.
- The target function has discrete output values
- Disjunctive descriptions may be required
- The training data may contain missing attribute values

if - else

can handle

Real world applications

- Biomedical applications
- Manufacturing
- Banking sector
- Make-Buy decisions

numerical ? + next class

Practice Exercises (for Students)



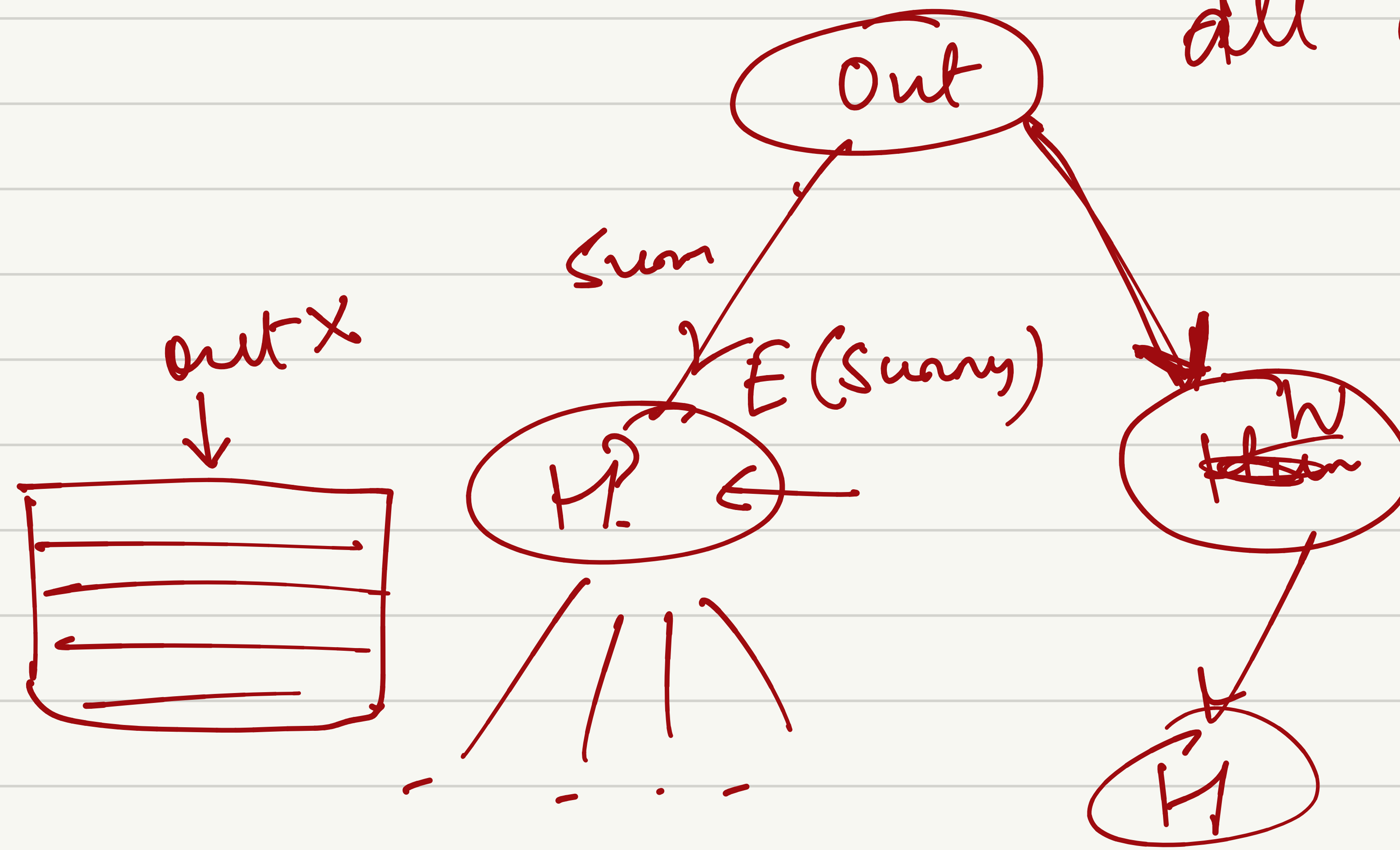
1. For this similar problem discussed in class build a decision tree classifier using ID3 algorithm ie., Information gain and entropy measures. Grow the complete decision tree.
2. Can "Object ID" be used as one of key feature to construct the tree? Justify your answer on the results of entropy calculated.
3. Use the test data to construct the confusion matrix and find the precision , recall, F-score.
4. Compare the training accuracy vs test accuracy for the model built in part 1)

EC2 - till CS8

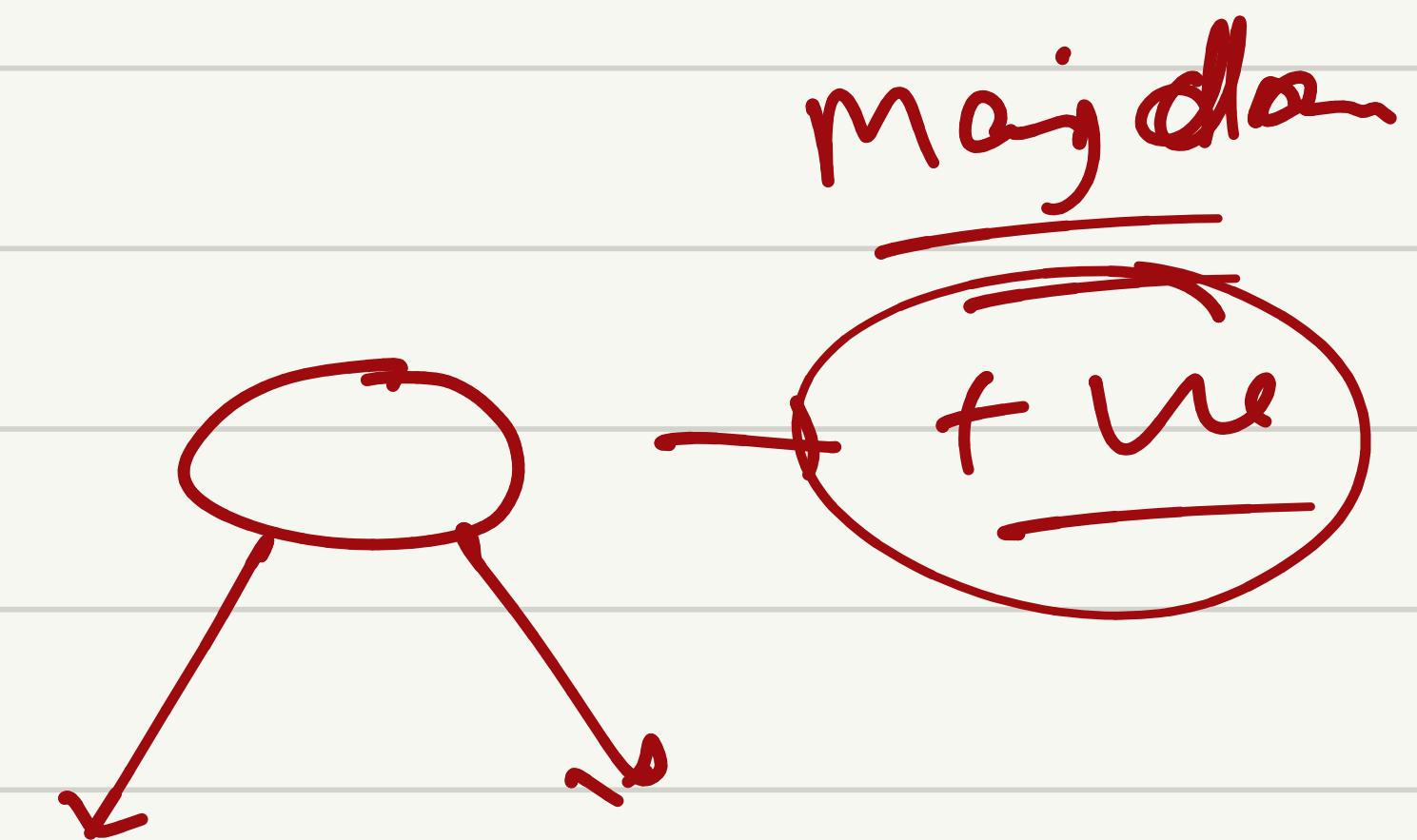
	Training Data			
Object ID	Shape	Colour	Size	Action
OB101	Round	Green	Small	Reject
OB102	Square	Black	Big	Allow
OB103	Square	Brown	Big	Allow
OB104	Round	Brown	Small	Reject
OB105	Square	Green	Big	Allow
OB105	Square	Brown	Small	Reject
OB106	Oval	Green	Big	Reject
OB107	Oval	Brown	Small	Allow
OB108	Oval	Green	Small	Reject

	Test Data / Prune Set/ Validation Set			
Object ID	Shape	Colour	Size	Action
OB109	Oval	Black	Small	Reject
OB110	Round	Brown	Big	Allow
OB111	Square	Brown	Big	Allow
OB112	Oval	Green	Small	Allow

all attribute



$E \rightarrow$
IG



Mid Term Examination

Important Note to students:

- Handout mapped prescribed book sections, class discussions , practice exercises shared in all the slides must be revised for the exam preparation
- Since the mid term exam is CLOSED BOOK mode, no reference material is allowed and hence formula needs to be learnt by the students
- Queries may not be answered once the exam starts.
- **Kindly Plan ahead to get your preparation queries resolved before the EXAM START DATE of your batch.**

Additional References



Decision Tree

- https://www.youtube.com/watch?v=eKD5gxPPeY0&list=PLBv09BD7ez_4temBw7vLA19p3tdQH6FYO&index=1



Thank you !